

1. (a) (i) For all $x \neq 1$, $X_n(x) = 0$ for large n , and $P(\{1\}) = 0$. This proves $X_n \rightarrow X$ almost surely (P) as $n \rightarrow \infty$.
- (ii) Since for fixed $T > 0$, $\int |X_n| I_{\{|X_n| > T\}} dP = n^2 \cdot m([1 - \frac{1}{n^2}, 1]) = n^2 \cdot \frac{1}{n^2} = 1$, $\forall n > \sqrt{T}$, $\{X_n\}$ is not UI.
- (iii) Here $E|X_n - X| = E(X_n) = n^2 \cdot \frac{1}{n^2} = 1$ for all n large, so X_n does not converge to X in L^1 .
- (b) If $\{X_n^p\}$ is UI, and the measure is finite (which is the case in the example above), then $X_n \rightarrow X$ a.e (P) would imply $X_n \rightarrow X$ in L^p for $p \geq 1$ (HW problem: 2.36(b)). There are other correct answers too, for example, along the lines of DCT.

2. (a) $\mu_3(\emptyset) = 3\mu_1(\emptyset) + 2\mu_2(\emptyset) = 0$; and $\mu_3(A) = 3\mu_1(A) + 2\mu_2(A) \geq 0 \Rightarrow \mu_3(A) \in [0, \infty]$, $\forall A \in \mathcal{F}$; If $\{A_i\}$ is a disjoint collection of sets in $\mathcal{F}$, then $\mu_3(\cup_i A_i) = 3\mu_1(\cup_i A_i) + 2\mu_2(\cup_i A_i) = 3\sum_i \mu_1(A_i) + 2\sum_i \mu_2(A_i) = \sum_i (3\mu_1(A_i) + 2\mu_2(A_i)) = \sum_i \mu_3(A_i)$. So μ_3 is a measure.

- (b) $\nu(A) = 0 \Rightarrow \mu_1(A) = \mu_2(A) = 0 \Rightarrow \mu_3(A) = 3\mu_1(A) + 2\mu_2(A) = 0$. So $\mu_3 \ll \nu$.

Will show: $\frac{d\mu_3}{d\nu} = 3\frac{d\mu_1}{d\nu} + 2\frac{d\mu_2}{d\nu}$ a.e (ν). Since $\mu_1 \ll \nu$ and $\mu_2 \ll \nu$, then $\exists f_1, f_2 \geq 0$ s.t $\mu_1(A) = \int_A f_1 d\nu$, $\mu_2(A) = \int_A f_2 d\nu$. By definition, $\mu_3(A) = 3\mu_1(A) + 2\mu_2(A) = 3\int_A f_1 d\nu + 2\int_A f_2 d\nu = \int_A (3f_1 + 2f_2) d\nu$ by linearity of integrals. Hence by uniqueness of RN derivatives, $\frac{d\mu_3}{d\nu} = 3f_1 + 2f_2 = 3\frac{d\mu_1}{d\nu} + 2\frac{d\mu_2}{d\nu}$ a.e (ν).

- (c) Let $B = \mathbb{N} \cup \{0\} = \mathbb{N}_0$, then $[(0.7)\mu_2](B^c) = 0 = \nu(B)$, i.e. $(0.7)\mu_2 \perp \nu$. Choose $\mu_s = (0.7)\mu_2$, $\mu_a = (0.3)\mu_1$, then $\mu = \mu_a + \mu_s$. To show $\mu_a \ll \nu$ and compute $\frac{d\mu_a}{d\nu}$. Note that if $\phi_{2,3}$ represent the density of $Normal(2, 3)$, then by definition $\mu_1(A) = \int_A \phi_{2,3} dm$ and $\nu = m$. So, $\frac{d\mu_1}{d\nu} = \phi_{2,3}$ from RN theorem. Hence, from the proposition in the text, $\frac{d\mu_a}{d\nu} = \frac{d((0.3)\mu_1)}{d\nu} = (0.3)\frac{d\mu_1}{d\nu} = (0.3)\phi_{2,3}$.

- (d) Using change of variable formula, we know:

$$E(X^2) = \int x^2 dP_X(x) = \int x^2 d\mu(x) = (0.3) \int x^2 d\mu_1(x) + (0.7) \int x^2 d\mu_2(x) = (0.3)(7) + (0.7)(42)$$

= 31.5, since second moment of $Normal(2, 3)$ is $(3 + 2^2) = 7$ and second moment of Poisson ($\lambda = 6$) is $\lambda + \lambda^2 = 6 + 6^2 = 42$

3. Let $f(X) = |X|^{p_1}$, $g = 1$, $r = \frac{p_2}{p_1} > 1$, $q = \frac{r}{r-1}$ where $\frac{1}{r} + \frac{1}{q} = 1$. Since $E|f|^r = E|X|^{p_2} < \infty$ and $E|g|^q = E1 = 1 < \infty$, then by Hölder's inequality, $E|fg| \leq (E|f|^r)^{\frac{1}{r}} (E|g|^q)^{\frac{1}{q}}$, i.e. $E(|X|^{p_1} \cdot 1) \leq [E(|X|^{p_2})]^{\frac{1}{r}} \cdot 1$, i.e. $E(|X|^{p_1}) \leq [E(|X|^{p_2})]^{\frac{p_1}{p_2}}$. Since $E|X|^{p_2} < \infty$, then the above inequality gives $E|X|^{p_1} < \infty$, i.e. $X \in L^{p_1}(\Omega, \mathcal{F}, \mu)$.